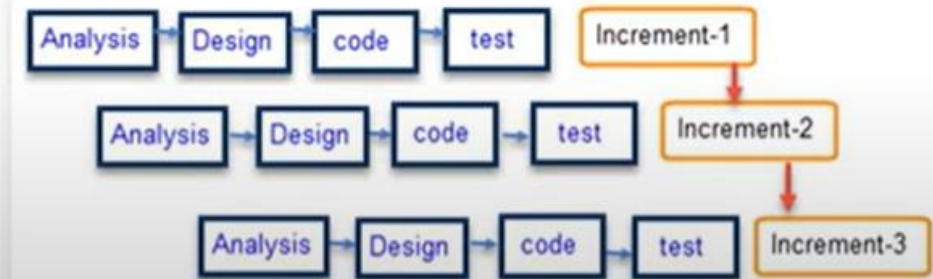


Incremental Process Models

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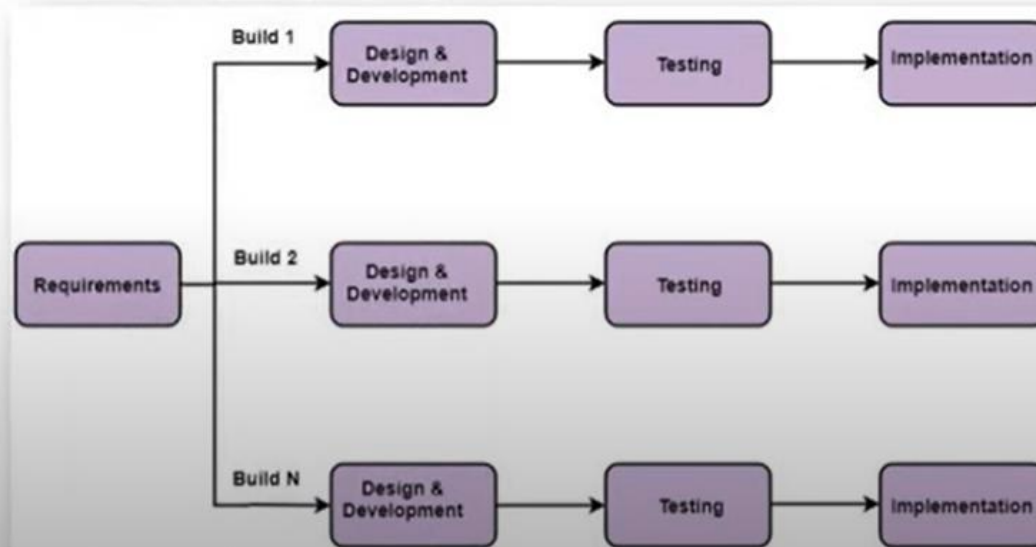


Incremental Model



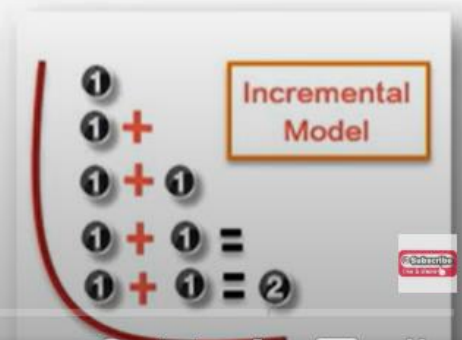
Incremental Process Model

- In Incremental Model, requirements divided into multiple standalone modules of the software development cycle.
- Each module goes through the requirements, design, implementation and testing phases.
- Every subsequent release of the module adds function to the previous release.
- The process continues until the complete system achieved.



When to use the Incremental process model

1. Requirements are clearly specified, understood and are known up-front.
2. When the customer demands a quick release of the product.
3. You can develop prioritized requirements first.
4. A project has a long development schedule.
5. When Software team are not very well skilled or trained.
6. Lowers initial delivery cost.
7. Such methodology is more in use for web application and product based companies



Incremental process model phases

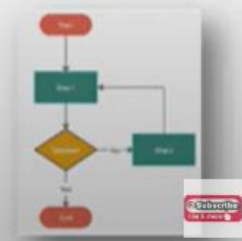
1. Requirement Gathering & Analysis:

- Business Analyst & Product Manager collect all the Functional & Non Functional requirements of customers.
- This phase performs a crucial role in Incremental Model.



2. Design Phase:

- It design the overall software architecture together with high level and detailed design.
- Designer design Algorithm, Data Flow diagram, Activity diagram, Class diagram, State Transition diagram, Database design etc.



Incremental process model phases

3. Implementaion Phase:

- In this phase software design is translated into specific source code.
- Programmers use suitable programming languages, frameworks & databases.



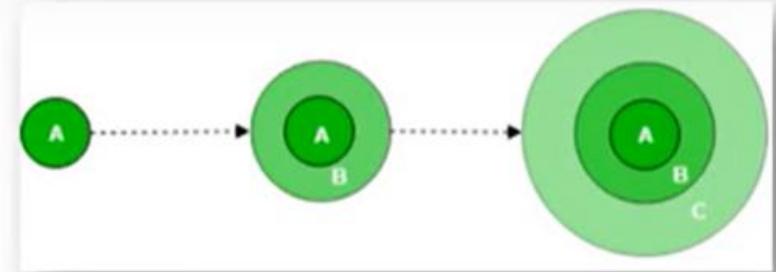
4. Testing Phase:

- In the testing phase, Tester checks the performance of each existing function as well as additional functionality as per customer requirements or not.
- There are many test methods, but the most common are white box, black box and grey box test methods.



Advantages of Incremental process models

1. The Client gets important functionality early.
2. Clients have a clear idea of the project.
3. Errors are easy to be recognized.
4. Easier to test and debug
5. Simple to manage risk because it handled during its iteration.
6. Changes are easy to implement.
7. It is flexible and less expensive to change requirements and scope.
8. Partial systems are successively built to produce a final total system.



Disadvantages of Incremental process model

1. Well defined module interfaces are needed.
2. A good team and proper planned execution are required because of its continuous iterations.
3. Problems might cause due to system architecture as such not all requirements collected up front for the entire software lifecycle.
4. Each iteration phase is rigid and does not overlap each other.

